PEDIATRIC INFECTIOUS DISEASE UNIT
FACULTY OF MEDICINE, CHIANG MAI UNIVERSITY

Invasive fungal infection

for pediatric residents

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Outlines

- Overview of fungal infection
- Invasive fungal infection
  - Yeast: Invasive candidiasis
  - Mold: Invasive aspergillosis, mucormycosis

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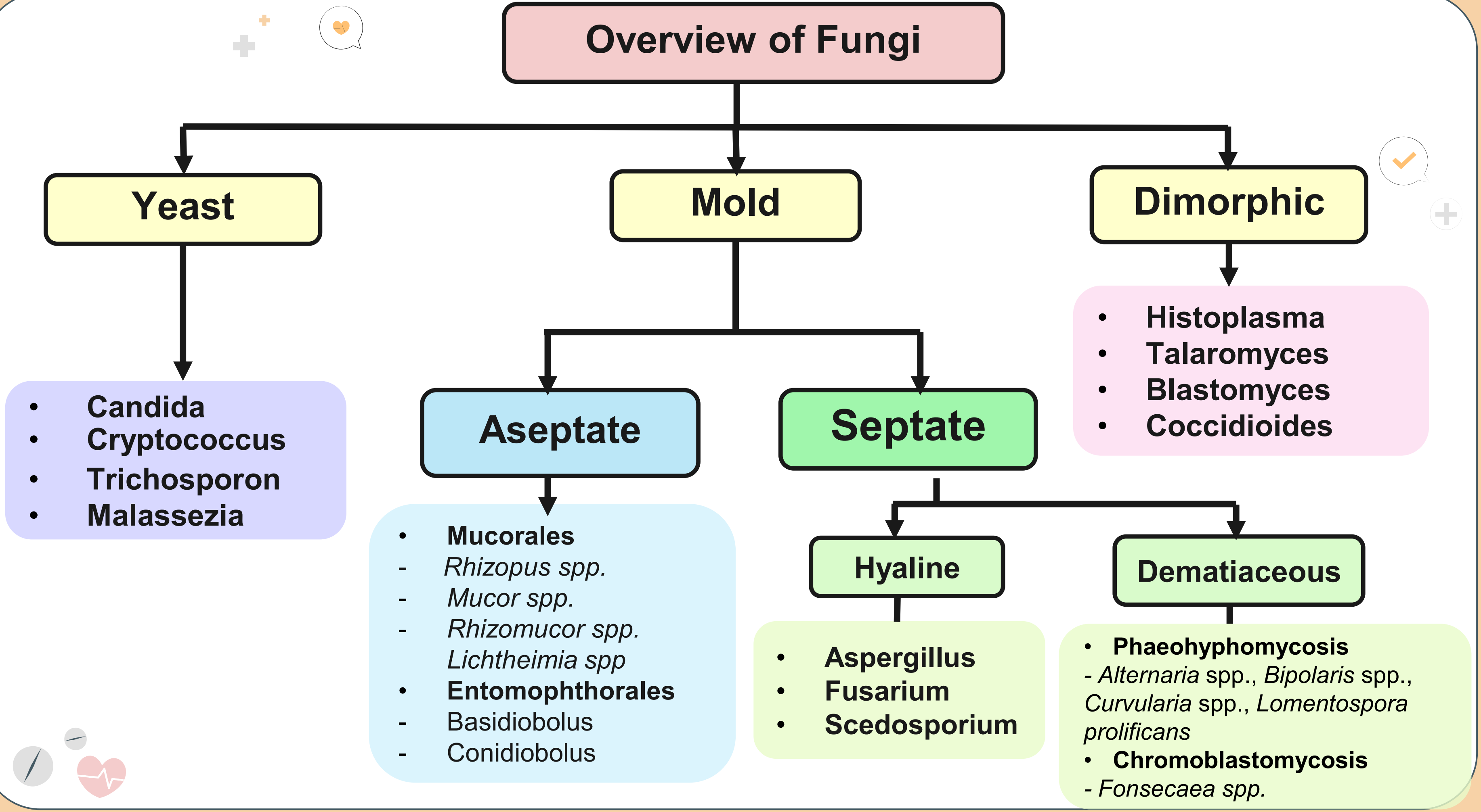
Overview of fungal infection



Overview of Fungi
Yeast
Mold
Dimorphic
• Candida
• Cryptococcus
• Trichosporon
• Malassezia
Aseptate
Septate
• Histoplasma
• Talaromyces
• Blastomyces
• Coccidioides
• Mucorales
- *Rhizopus spp.*
- *Mucor spp.*
- *Rhizomucor spp.*
Lichtheimia spp
• Entomophthorales
- Basidiobolus
- Conidiobolus
Hyaline
Dematiaceous
• Aspergillus
• Fusarium
• Scedosporium
• Phaeohyphomycosis
- *Alternaria* spp., *Bipolaris* spp., *Curvularia* spp., *Lomentospora prolificans*
• Chromoblastomycosis
- *Fonsecaea spp.*

2

Invasive candidiasis

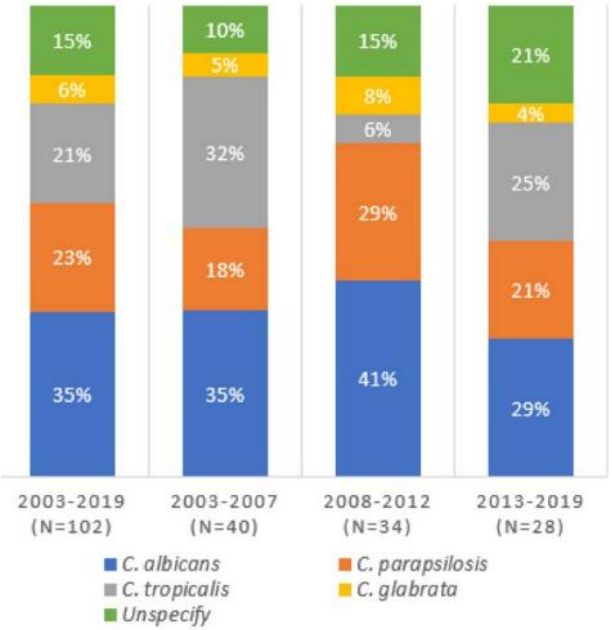
Introduction

- Candidiasis encompasses many clinical syndromes that may be caused by several species of *Candida*
- Invasive candidiasis
  - *Candida* infections of the blood and other sterile body fluids
  - Leading cause of infection-related mortality in hospitalized immunocompromised patients.

Nelson textbook of pediatrics, 22nd edition

Epidemiology

- Species distribution of pediatrics IC during 2003–2019[1]



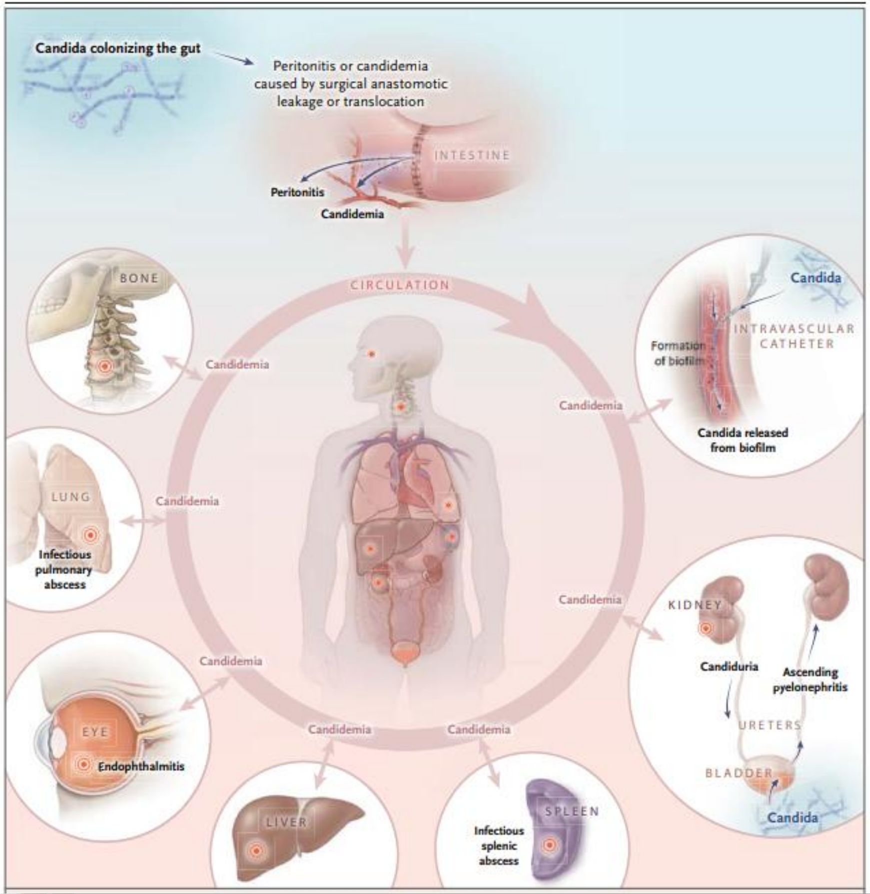
Common candida species[2]

- *C. albicans* (most human infections)
- *C. parapsilosis*
- *C. tropicalis*
- *C. krusei*
- *C. lusitaniae*
- *C. glabrata*
- *C. auris* → emerging multiresistant invasive pathogen, immunocompromised host

C. non-albicans

*Amphotericin B →inactive against 20% of *C. lusitaniae*
**Fluconazole resistance[2,3]
- Most strains of *C. krusei*
- 5–50% of *C. glabrata* isolates
- 90% of *C. auris* isolates

1. Watsamon J, et al. The Pediatric Infectious Disease Journal 40(2):p 96-102, February 2021
2. Nelson textbook of pediatrics, 22nd edition
3. Redbook, 33rd edition, 2024-2027



Pathogenesis of Invasive Candidiasis

Overgrowth ← Modified microbiota

↓

Mucosal colonization

Oro-pharyngeal
Upper + lower digestive tract
Genital tract
Urinary tract

↑ Diabetes
Burn
Neutropenia
Antibiotics
Prematurity

Micro-invasion ← Candidemia

Multiple antibiotics
Vascular accesses
Parenteral nutrition
ICU stay > 7 days
Candida colonization
Renal failure
Major abdominal surgery

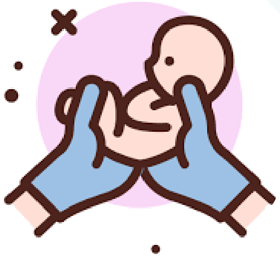
↑ Endopthalmitis
Endocarditis
Catheter-related
Abscess
CNS
Hepato-splenic

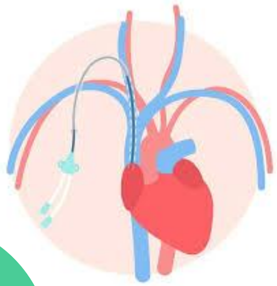
Candidemia → Disseminated disease

Eggimann et al. Annals of Intensive Care 2011, 1:37
Kullberg BJ, Arendrup MC. Invasive Candidiasis. N Engl J Med. 2015

Risk factors for invasive candidiasis

- Critical illness, long-term ICU stay
- Abdominal surgery → Anastomotic leakage or repeat laparotomies
- Acute necrotizing pancreatitis
- Hematologic malignant disease
- Solid-organ transplantation, solid-organ tumors
- Neonates, particularly those with low birth weight, and preterm infants
- Use of broad-spectrum antibiotics
- Presence of central vascular catheter, total parenteral nutrition (TPN)
- Hemodialysis
- Glucocorticoid use or chemotherapy





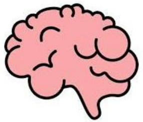
Kullberg BJ, et al. N Engl J Med 2015;373:1445-1456

Clinical manifestations

❑ Neonatal invasive candidiasis

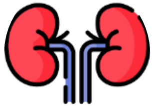
- Overwhelming sepsis
- **Signs of invasive candidiasis among premature infants** (non-specific)
    - Temperature instability
    - Lethargy
    - Apnea
    - Hypotension
    - Respiratory distress
    - Abdominal distention
    - Thrombocytopenia (>80%)

Complications:



CNS involvement: meningoencephalitis, abscesses
- unremarkable CSF parameters (WBC, glucose, protein)
- 50% of candida meningitis -> not have blood culture positive

Endophthalmitis (<5%)



Renal involvement
- Candiduria, diffuse infiltration in renal parenchyma, collecting system

Others: Heart, bones, joints, liver, and spleen

Candiduria should be considered a **surrogate marker of candidemia** in premature infants.

Nelson textbook of pediatrics, 22nd edition

Neonatal infections

- Cumulative incidence of candidiasis among infants <1,000 g birthweight ranges from 2% to 28%.
  - Colonization rates increase to >50% among infants admitted to the NICU
- **Significant risk factors for neonatal invasive candidiasis**
  - Prematurity, low birthweight
  - Exposure to broad-spectrum antibiotics
  - Abdominal surgery
  - Endotracheal intubation
  - Presence of a central venous catheter
- **Pathogenesis**
  - Immunologic immaturity along with an underdeveloped layer of skin, need for invasive measures (endotracheal tubes, central venous catheters)
  - Decreasing the physiologic barriers

Nelson textbook of pediatrics, 22nd edition

Clinical manifestations

❑ Cancer and transplant patients

- Fever > 5 days of during neutropenic episode
- **High-risk oncology patients**
  - **Prophylaxis:** fluconazole, echinocandins
- **Bone marrow transplant recipients**
  - Higher risk of fungal infections due to prolonged duration of neutropenia
  - **Prophylaxis:** voriconazole (benefit over fluconazole of mold prophylaxis)
- **Solid organ transplant recipients**
  - **Prophylaxis in high risk** (prolonged surgical time, comorbidities, recent antibiotic exposure, or bile leak)
    - Amphotericin B, fluconazole, voriconazole, or caspofungin

❑ Catheter-associated infections

- **Risk for *Candida* spp. central catheter infection**
  - Neutropenia
  - Use of broad-spectrum antibiotics,
  - TPN

Nelson textbook of pediatrics, 22nd edition

Diagnosis for invasive candidiasis

Culture-based methods

- **Positive blood cultures for *candida* spp.**
  - ≥90% of positive cultures identified within 72 hrs
  - Sensitivity 21–71%[1,2], specificity NA
  - **Optimal volume:** 3-5 ml (children)
  - **Interpretation:** negative results of culture do not exclude invasive infection in immunocompromised hosts

Nonculture-based methods

- **Serum (1,3)-β- D -glucan (for high risk)**
  - polysaccharide component of the fungal cell wall
- **Candida antigen and anti-Candida antibody detection**
- **PCR sequencing DNA-based method: 18s rRNA**
- **T2Candida Panel**

Definitive diagnosis of invasive candidiasis:

- Isolation of the organism from a normal sterile body site eg. **blood, CSF, bone marrow**), or
- Tissue biopsy specimen

1. Kullberg BJ, et al. N Engl J Med 2015;373:1445-1456
2. Redbook, 33rd edition, 2024- 2027

Management

- **Asymptomatic candiduria**
  - Elimination of predisposing factors, such as indwelling bladder catheters
  - Antifungal treatment is not recommended unless patients are at high risk of candidemia (neutropenic patients and preterm infants)

Candiduria

→ **Preterm infant**

→ **Evaluate and treatment**

- Blood cultures
- Lumbar puncture
- Ophthalmologic examination
- Brain imaging
- Abdominal ultrasonography

→ **Term infant**

- Need only ultrasound of renal system

Redbook, 33rd edition, 2024-2027

Management

Neonatal invasive candidiasis

- Remove or replace central line once the diagnosis of candidemia
  - Delayed removal → ↑ mortality and morbidity, poor neurodevelopmental outcomes
- **Duration of treatment:**
  - If candidemia without obvious metastatic complications: **2 weeks after documented clearance of *Candida* species from the bloodstream**
- Antifungal therapy should be targeted based on susceptibility testing.
- **Treatment for systemic candidiasis**
  - **First line:**
    - **Amphotericin B deoxycholate**
      - Better tolerated in infants than in adult patients
  - **Second line:**
    - Fluconazole: useful for UTI, with high concentrations in urine
    - Echinocandins: for failure treatment
    - Liposomal amphotericin
      - Worse outcomes in infants, used only when urinary tract involvement be excluded

Prognosis

- Mortality in premature infants: 20%- 50% in infants BBW <1,500 gm
- Associated with poor neurodevelopmental outcomes, chronic lung disease, and severe retinopathy of prematurity

Nelson textbook of pediatrics, 22nd edition
Pappas PG, et al. Clin Infect Dis 2016;62(4):e1-50

Further investigations

Further assessment of infants in the **presence of documented candidemia**

- **Ultrasound or CT of the head**
  - To evaluate for abscess
- **Ultrasound abdomen:** liver, kidney, and spleen
- **Echocardiogram**
- **Ophthalmologic exam:** to evaluate endophthalmitis
- **Lumbar puncture**
- **Urine culture for fungus**

Nelson textbook of pediatrics, 22nd edition

Management

Invasive candidiasis (beyond the neonatal period)

Empirical therapy for moderately or severely ill children with neutropenia

- **Preferred treatment:** **Echinocandins** (caspofungin, micafungin, and anidulafungin)
  - Activity against most *Candida* species
- **Alternative treatment:**
  - Fluconazole (if clinically stable and unlikely to have fluconazole resistance)
  - Amphotericin B
- Definitive antifungal selection based on susceptibility testing results
- Removal of infected devices (vascular or peritoneal catheters ventriculostomy drains shunts

Fungal Species	Echi-nocandins[a]
Candida albicans	++
Candida tropicalis	++
Candida parapsilosis	+
Candida glabrata	+/-
Candida krusei	++
Candida lusitaniae	+
Candida guilliermondii	+/-
Candida auris	++

Redbook, 33rd edition, 2024-2027

Anti-fungal drugs

Fungal Species	Amphotericin B Formulations	Flucona-zole	Itraconazole	Voriconazole	Posacona-zole	Isavucona-zole	Flucytosine	Echi-nocandins[a]
Candida albicans	+	++	+	++	+	+	+	++
Candida tropicalis	+	++	+	++	+	+	+	++
Candida parapsilosis	++	++	+	++	+	+	+	+
Candida glabrata	+	–	–	–	+/-	+/-	+	+/-
Candida krusei	+	–	–	+	+	+	+	++
Candida lusitaniae	–	++	+	++	+	+	+	+
Candida guilliermondii	+	+	+	+	+	+	+	+/-
Candida auris	+/-	–	+/-	+/-	+	+	+/-	++

Redbook, 33rd edition, 2024-2027

Dosing of antifungal agents for invasive candidiasis

Drug	Pk studied in infants	Infants	Older than 1 year of age
Amphotericin B deoxycholate	Yes	1 mg/kg/day	1 mg/kg/day
Amphotericin B lipid complex	Yes	5 mg/kg/day	5 mg/kg/day
Liposomal amphotericin B	Yes	5 mg/kg/day	5 mg/kg/day
Fluconazole	Yes	12 mg/kg/day	12 mg/kg/day loading then 6 mg/kg/day
Voriconazole	No	-	8 mg/kg every 12 hr
Micafungin	Yes	10 mg/kg/day	2-4 mg/kg/day
Caspofungin	Yes	50 mg/m^2/day	50 mg/m^2/day
Anidulafungin	Yes	1.5 mg/kg/da	1.5 mg/kg/day

Adapted from Nelson textbook of pediatrics, 22nd edition

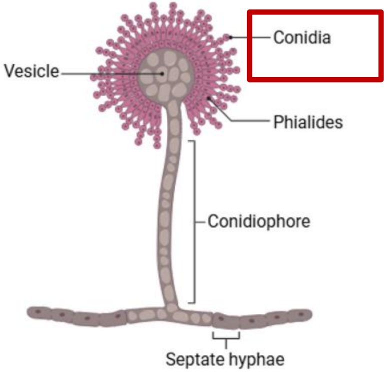
Prophylaxis

- **NICUs with a high incidence (>10%) of invasive candidiasis**[1]
    - Consider prophylaxis with fluconazole in infants BBW <1,000 g
        - Reducing invasive candidiasis
- **Drug and doses**[1,2]
    - **Fluconazole 3 or 6 mg/kg/dose, twice-weekly for 6 weeks**[1]
        - Decreases rates of both colonization and invasive fungal infections
        - Not increase the frequency of infections caused by fluconazole-resistant strains

1. Clinical Infectious Diseases, Volume 62, Issue 4, 15 February 2016, https://doi.org/10.1093/cid/civ933
2. Nelson textbook of pediatrics, 22nd edition

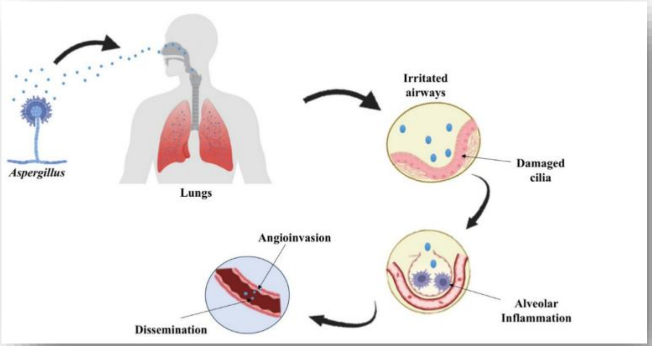
3 Invasive aspergillosis

Epidemiology



- Genus *Aspergillus* contains approximately 250 species
- **Route of transmission:**
  - Inhalation of conidia (spores) from environmental sources (e.g. plant, vegetables, dust from construction), soil and water supplies → subsequently germinate into fungal hyphae and invade host tissue

- Germination occurs throughout the immune response in immunocompromised individuals
  - insufficient immune response fails to impede conidial development



Redbook, 33rd edition, 2024-2027
Singh A, et al. J Food Drug Anal. 2025 Jun.

Aspergillosis

3 major forms of aspergillosis

Allergic disease (Hypersensitivity syndromes)

- Asthma
- Allergic bronchopulmonary aspergillosis (ABPA)*
- Allergic aspergillus sinusitis*

Saprophytic (noninvasive) syndromes

- Pulmonary aspergilloma*
- Chronic pulmonary aspergillosis*
- Otomycosis

Invasive aspergillosis*

* Principal clinical entities

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Allergic bronchopulmonary aspergillosis (ABPA)

- Hypersensitivity disease resulting from immunologic sensitization to *Aspergillus* antigens.
- Common: asthma, cystic fibrosis
- Clinical manifestation:
  - Wheezing, pulmonary infiltrates, bronchiectasis, and fibrosis
- **Criteria**

Primary diagnostic criteria:

- Episodic bronchial obstruction
- Peripheral eosinophilia
- Immediate cutaneous reactivity to Aspergillus antigens, precipitating IgE antibodies to Aspergillus antigen
- Elevated total IgE, serum precipitin (specific IgG) antibodies to *A. fumigatus*
- Pulmonary infiltrates, central bronchiectasis

Secondary diagnostic criteria:

- Repeated detection of *Aspergillus* from sputum by identification of morphologically consistent fungal elements or direct culture and coughing of brown plugs or specks.
- Radiology: bronchial wall thickening, pulmonary infiltrates, and central bronchiectasis

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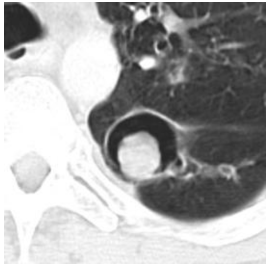
Pulmonary aspergilloma

- Aspergillomas
  - masses of fungal hyphae, cellular debris, and inflammatory cells → proliferate without vascular invasion
- Preexisting cavitary lesions: TB, histoplasmosis, resolved abscesses pulmonary cysts, bullous emphysema.
- **Clinical:** may be asymptomatic, hemoptysis, cough, or fever
- **Imaging:**
  - Thickening of the walls of a cavity, and later on there is a solid round mass separated from the cavity wall as the fungal ball develops
- **Diagnosis:** Detection of *Aspergillus* antibody in the serum
- **Treatment**
  - Systemic antifungal treatment with azole-class agents (if indicated)
  - **Definitive treatment:** surgical resection



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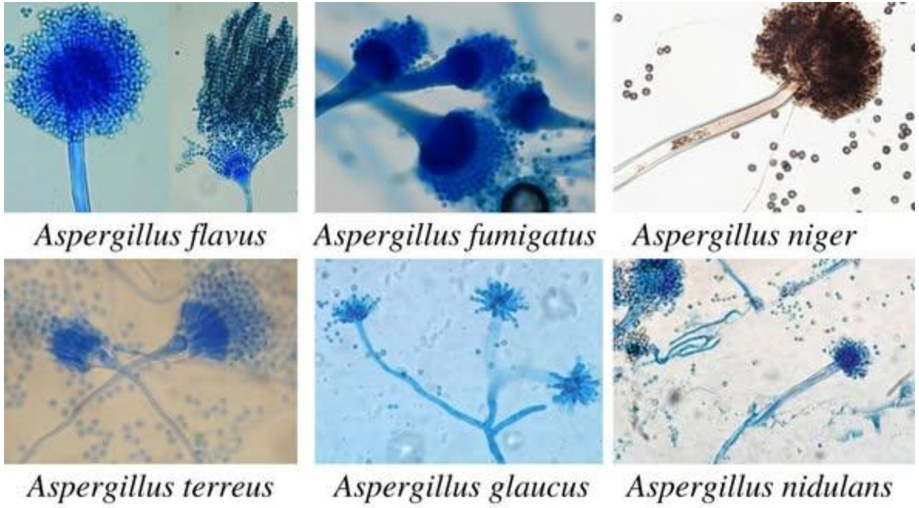
Sarah S, Long. et al, Principles and Practice of Pediatric Infectious Diseases, 6th edition

Invasive aspergillosis



Etiology:

- ***Aspergillus fumigatus* (>75%)**
- ***Aspergillus flavus***
- Other major species
  - *Aspergillus terreus:* intrinsic resistance to amphotericin B
  - *Aspergillus nidulans* (common in CGD)
  - *Aspergillus niger*

Redbook, 33rd edition, 2024-2027

Invasive aspergillosis

- Organ involvement: pulmonary, sinus, cerebral, cutaneous site

- **Invasive pulmonary aspergillosis (IPA)** = most common form of aspergillosis
- Rare: IE, osteomyelitis, meningitis, peritonitis, infection of the eye or orbit

Host factors:

- Immunocompromised host with prolonged neutropenia
- Graft-versus-host disease
- Impaired phagocyte function (e.g. CGD)
- Received T-lymphocyte immunosuppressive therapy
- Steroid
- Calcineurin inhibitors; tacrolimus
- TNF-alpha inhibitors; infliximab etanercept

Children with highest risk:

- New-onset AML
- Relapse of hematologic malignancy
- Aplastic anemia
- CGD
- Recipients of allogeneic HSCT, SOT (lungs)

Redbook, 33rd edition, 2024-2027
Nelson textbook of pediatrics, 22nd edition

Clinical manifestations

- **Hallmark of invasive aspergillosis**
  - Angioinvasion with thrombosis
  - Dissemination to other organs
  - Erosion of the blood vessel wall with catastrophic hemorrhage

- **Clinical presentation of IPA:**
  - Fever despite initiation of empirical broad spectrum antibacterial therapy
  - Cough, chest pain, hemoptysis, and pulmonary infiltrates

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Nelson textbook of pediatrics, 22nd edition

Diagnostic investigations for IPA

1. **Culture-based methods:**
   - Culture of *Aspergillus* from a normally sterile site and histologic identification of tissue invasion by fungal hyphae consistent with *Aspergillus* morphology
2. **Imaging: CT chest with contrast**
3. **Galactomannan assay**
   - Specific for Aspergillus
   - Specimen: serum or from bronchoalveolar lavage (BAL)
4. **(1,3)-β-D-glucan assay**
   - Nonspecific molecular fungal assay
   - Detect the major component of the fungal cell wall
5. **Polymerase chain reaction (PCR)** based assays: 18srRNA

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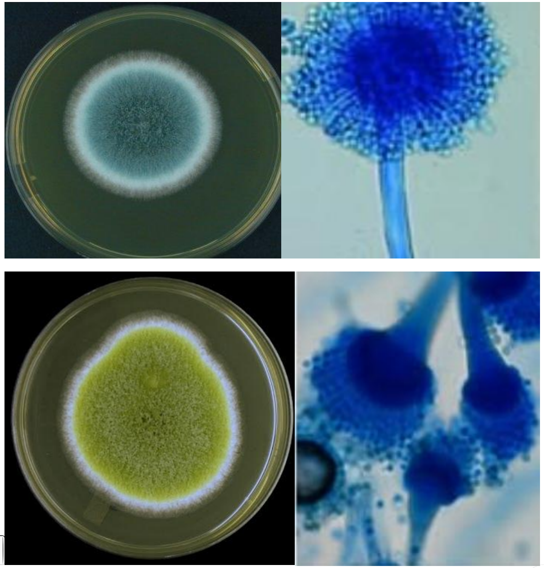
1. *Aspergillus* spp. identification

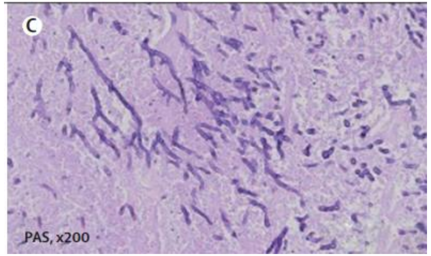
- **Definitive diagnosis:** require isolation of mold from culture in addition to histologic demonstration
- **Gold standard:** tissue biopsy

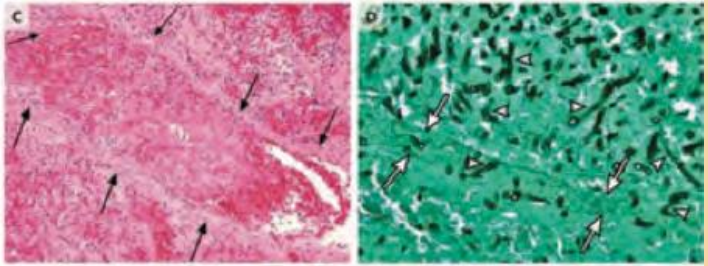
Culture-based methods



Histopathologic examination:

- Dichotomous branching with septate hyphae
- Acute-angle branching (approximately 45°)
- Hyphae are generally 2.5–4.5 μm in width, display parallel walls,
- Often invading tissue → Vascular invasion, necrosis, and a surrounding inflammatory response
- Regular pattern

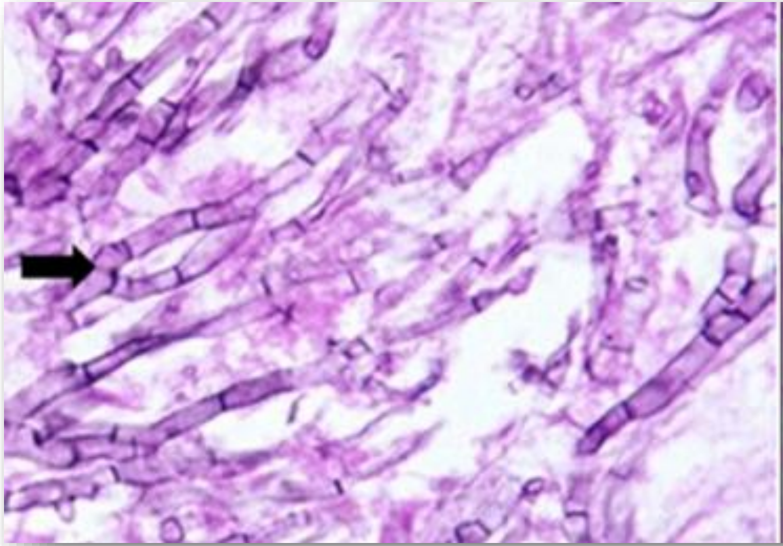




Grocott-Gomori methenaine silver nitrate stain (GMS), with hyphal walls staining dark

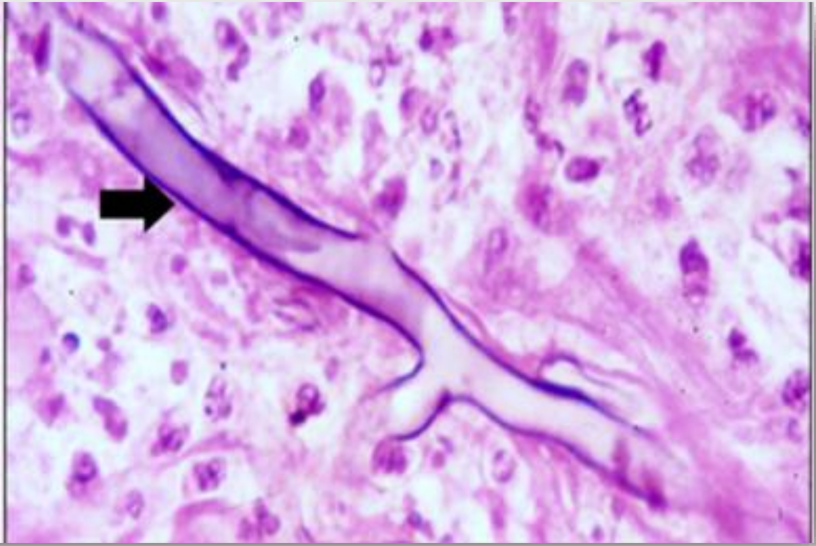
Redbook Atlas of pediatric infectious disease. 4th edition. 2020
Douglas AP, et al. Intern Med J. 2021 Nov;51 Suppl 7:143-176
Sarah long, et al. Principles and Practice of Pediatric Infectious Diseases, 6th edition

Aspergillus spp.



- Narrow, septate hyphae
- Acute angled branching
- Regular pattern

Mucorales



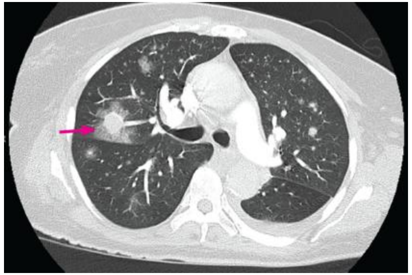
- Broad, Pauci-septate or aseptate hyphae, ribbon-like hyphae
- Right angled branching
- Irregular pattern

2. CT chest with contrast in IPA

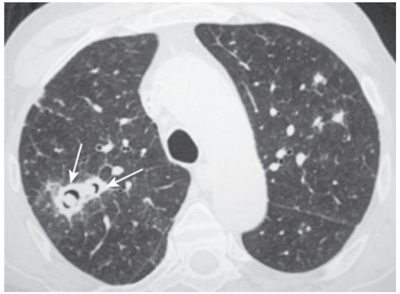
- **Imaging: recommended CT with contrast** (nodule or a mass is close to a large vessel)

Classic radiologic signs with IPA

- **Halo sign** (Rim of ground glass opacity surrounding the nodule)
- **Air crescent sign** (Crescent of gas between an intra-cavitary mass (may be due to fungal ball) and the cavity wall
- **Cavity**
- **Wedge-shaped and segmental or lobar consolidation**



Halo sign
(during neutropenia)



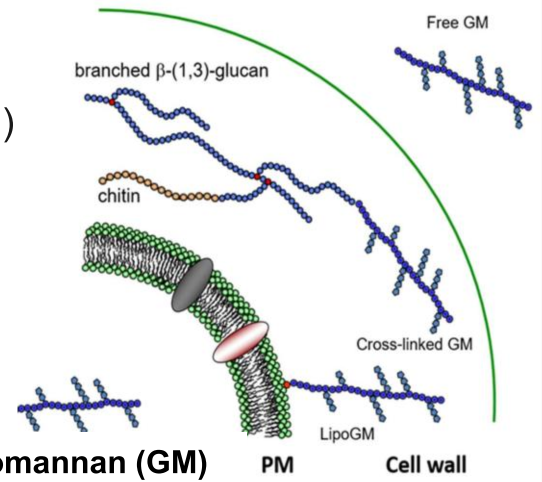
Consolidation with an eccentric cavitation and **air crescent sign** (during recovery)

Nelson textbook of pediatrics, 22nd edition
Clinical Infectious Diseases, September 2020, Pages 1367–1376, https://doi.org/10.1093/cid/ciz1008

3. Galactomannan (GM) assays

- GM is molecule found in cell wall of *Aspergillus* spp.
- Specimen: serum, BAL fluid (recommend if suspicion of IPA) → useful in hematologic malignancy, HSCT (SOT→ poor sensitivity)

Cut-off (support diagnosis of IA)
Single serum or plasma ≥ 0.5
BAL fluid ≥ 1.0
Single serum or plasma ≥ 0.7 and BAL fluid ≥ 0.8



Galactomannan (GM)

Cautions:

- Negative GM test results does not exclude diagnosis of IA
- **False negative GM:** CGD
- **False positive GM:** *Fusarium, Penicillium, Histoplasma capsulatum,* colonization of gut neonates with Bifidobacterium species
- Not recommend GM for routine screening if receiving mold-active antifungal therapy or prophylaxis

Redbook, 33rd edition, 2024-2027
Douglas AP, et al. Intern Med J. 2021 Nov;51 Suppl 7:143-176

4. (1,3)-β-D-glucan assay

(1,3)-β-D-glucan assay

- Integral cell wall component, not released from fungal cell
- More non-specific than GM assay
- Not discriminate which fungal organism

Detection of several fungi:

- *Aspergillus* spp.
- *Candida* spp.
- *Fusarium* spp.
- *Trichosporon* spp.
- *Saccharomyces cerevisiae*
- *Acremonium*
- *Histoplasma capsulatum*
- *Sporothrix schenckii*
- *Pneumocystis jirovecii*

Except: *Cyptococcosis* spp., *Blastomyces* spp., *Mucor* spp., *Rhizopus* spp.

Sarah long, et al. Principles and Practice of Pediatric Infectious Diseases, 6th edition

Diagnostic criteria for invasive aspergillosis

Diagnosis	Criteria
Certain Histological/ culture + clinical/ radiologic abnormalities	Histological or cytological evaluation of lung tissue with hyphae on needle aspiration or biopsy in which hyphae or melanized yeast-like forms are associated to tissue damage *or* Positive culture test for Aspergillus on pulmonary specimen taken by sterile procedure *and* Clinical or radiologic abnormalities consistent with infection
Probable ≥1 host factor + culture/ PCR + clinical	At least 1 host factor (tab. 2) *and* Mycological evidence - positive microscopy or culture for *Aspergillus* on sputum, BAL bronchial brush, or aspirate - positive *Aspergillus* PCR (at least 2 tests) - positive antigenic assay[a] *AND* Clinical criteria consistent with infection[b]
Possible ≥1 host factor + clinical	At least 1 host factor (tab.2) *and* Clinical criteria compatible with infection[b]

[a] Positive antigenic assay: detection of galactomannan in plasma, serum, BAL, or CSF. β-D-glucan was not considered to provide mycological evidence of any invasive mold disease

[b] Clinical criteria compatible with infection: characteristic infiltrates on CT (dense, well-circumscribed lesions with or without halo sign, air crescent sign, or cavity), tracheobronchitis diagnosed by bronchoscopy, or infiltrates that are uncharacteristic but associated with specific pulmonary symptoms or signs (e.g., pleural pain, haemoptysis)

Abbreviations: *BAL* Bronchoalveolar lavage, *CSF* Cerebrospinal fluid, *CT* Computed tomography, *PCR* polymerase chain reaction

Terlizzi, V., *et al.* Management of pulmonary aspergillosis in children: a systematic review. *Ital J Pediatr* 49, 39 (2023)

Treatment of IPA in children

Medication	Dosage	Cautions
First-line		
Voriconazole (≥2 yrs)	Depending on age and body weight (kg)	- Not approved for children < 2 yrs - TDM strongly recommended
Alternative options		
Liposomal amphotericin B	IV: 3 mg/kg daily	- First line in < 2 yrs - Amphotericin B: preferred for neonates
Amphotericin B deoxycholate	IV 1-1.5 mg/kg daily	Used in neonates
Micafungin	< 50 kg: IV 2-4 mg/kg daily ≥ 50 kg: IV 100-200 mg daily	

- **Voriconazole = drug of choice for all clinical forms of IPA**
- **Except in;**
  - **Neonate**; High dose AmB deoxycholate is recommended because of voriconazole's potential visual adverse effects
  - Echinocandin can be used in the settings in which azole or polyene antifungals are contraindicated (not recommend for primary therapy)

TDM; Therapeutic Drug Monitoring:

Redbook, 33rd edition, 2024-2027
Nelson textbook of pediatrics, 22nd edition
Adapted from Douglas AP, et al. Intern Med J. 2021 Nov;51 Suppl 7:143-176.

Dosage for voriconazole in children and adolescents

Group	Dose q 12 hr		
	Loading dose	Maintainance dose	
	IV	IV	Oral
Children (2-12 yr) and young adolescents (12 -14 yr , BW < 50 kg)	9 mg/kg	8 mg/kg	9 mg/kg (max 350 mg)
Other adolescents (12–14 yr, BW > 50 kg and 15–16 years old) and adults	6 mg/kg	4 mg/kg	200 mg

- Oral bioavailability 96% in the fasting state (1 hr before or 1 hr after a meal), decreased by ~ 22% if taken with food
- * IV form is contraindicated if CrCl < 50 ml/min (cyclodextrin accumulation)

Redbook. 33rd edition. 2024
Friberg LE, et al. Antimicrob Agents Chemother 2012;56(6):3032-42.

Recommendations for antifungal TDM

Antifungal agent	Timing of first sample	Target serum concentration range (mg/L)	Published guidance for dose adjustment for TDM
Voriconazole	On day 3 or later	• Prophylaxis: 1–5.5 • Treatment: 2-6 • CNS infection, bulky disease, multifocal infection: >2	For crude adjustment method if trough concentration: • **0.0–0.5 mg/L**: increase dose by 50% • **>0.5–<1 mg/L**: increase dose by 25% • **>5.5 mg/L and asymptomatic**: decrease dose by 25% • **>5.5 mg/L with drug-related toxicities**: hold one dose and decrease subsequent doses by 50%

Voriconazole

- **Hepatic clearance**
- **Side effect:** Hepatotoxicity, increased QTc, CNS effects (associated with through level above 6 μg/mL), vision changes, phototoxicity
- Drug interactions or genetic polymorphisms involving P450 CYP2C19

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Drug interaction with voriconazole (1)

Effect of other drugs on voriconazole pharmacokinetics

Drug/Drug Class (Mechanism of Interaction by the Drug)	Voriconazole Plasma Exposure	Recommendation
Rifampin (CYP450 Induction)	Significantly Reduced	**Contraindicated**
Carbamazepine, phenobarbital (CYP450 Induction)	Likely to Result in Significant Reduction	**Contraindicated**
Phenytoin (CYP450 Induction)	Significantly Reduced	Increase voriconazole maintenance dose
Letermovir (CYP2C9/2C19 Induction)	Reduced	Monitor for reduced effectiveness of voriconazole
Fluconazole (CYP2C9, CYP2C19 and CYP3A4 Inhibition)	Significantly Increased	Avoid concomitant administration

Drug interaction with voriconazole (2)

Effect of voriconazole on pharmacokinetics of other Drugs

Drug/Drug Class (Mechanism of Interaction by the Drug)	Drug Plasma Exposure	Recommendation
Sirolimus (CYP3A4 Inhibition)	Significantly increased	**Contraindicated**
Cyclosporine (CYP3A4 Inhibition)	Significantly increased	Reduce the cyclosporine dose to one-half of the starting dose and follow with frequent monitoring of cyclosporine blood levels.
Tacrolimus (CYP3A4 Inhibition)	Significantly increased	Reduce the tacrolimus dose to one-third of the starting dose and follow with frequent monitoring of tacrolimus blood levels.
Omeprazole (CYP2C19/3A4 Inhibition)	Significantly increased	Reduce the omeprazole dose by one-half
Corticosteroids (CYP3A4 Inhibition)	Drug Exposure Likely to be Increased	Monitor for potential adrenal dysfunction

Spectrum of anti-fungal agents

Fungal Species	Amphotericin B Formulations	Flucona-zole	Itraconazole	Voriconazole	Posacona-zole	Isavucona-zole	Flucytosine	Echi-nocandins[a]
Candida albicans	+	++	+	++	+	+	+	++
Candida tropicalis	+	++	+	++	+	+	+	++
Candida parapsilosis	++	++	+	++	+	+	+	+
Candida glabrata	+	–	–	–	+/-	+/-	+	+/-
Candida krusei	+	–	–	+	+	+	+	++
Candida lusitaniae	–	++	+	++	+	+	+	+
Candida guilliermondii	+	+	+	+	+	+	+	+/-
Candida auris	+/-	–	+/-	+/-	+	+	+/-	++
Cryptococcus species	++	++	+	+	+	+	++	–
Trichosporon species	+	+	+	++	+	+	–	–
Aspergillus fumigatus[b]	+	–	+	++	+	++	–	+
Aspergillus terreus[b]	–	–	+	++	+	++	–	+
Aspergillus calidoustus[b]	++	–	–	–	–	–	–	++
Fusarium species[b]	+	–	–	++	+	+	–	–

Redbook. 33rd edition. 2024

Follow-up

- **Follow-up chest CT scan** to assess the response of IPA to treatment **after a minimum of 2 weeks of treatment**; earlier assessment is indicated if the patient clinically deteriorates

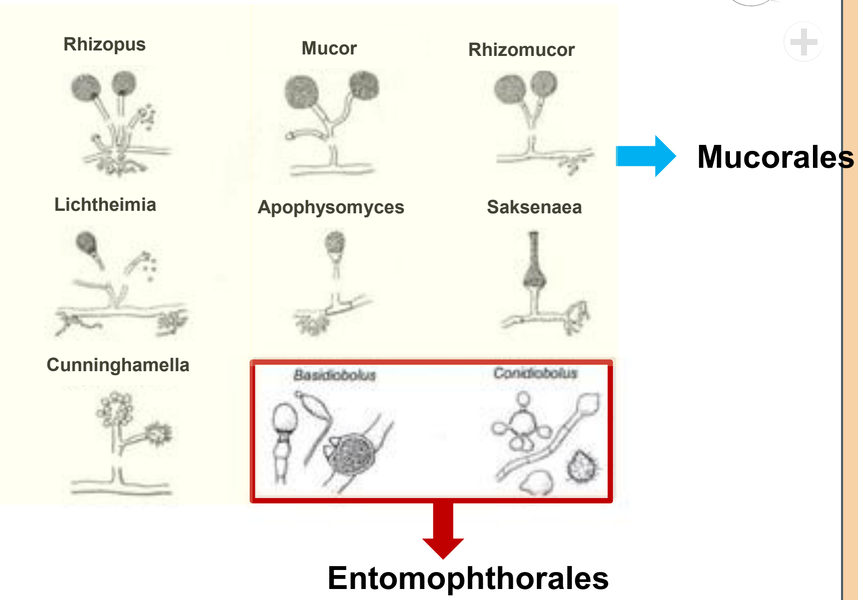
- **Duration of treatment: Generally minimum of 6–12 weeks**, depending on the severity, immunosuppression, and resolution of clinical disease (clinical and serial CT)
- Respond to therapy may be assessed by monitoring of serum galactomannan concentration (if significant elevated at onset)

Redbook. 33rd edition. 2024
Patterson TF, et al. Clin Infect Dis 2016;63(4):e1-e60.

4 Mucormycosis

Mucormycosis

- Mucormycosis is a very aggressive invasive fungal disease and life-threatening infection
- 3rd most common invasive fungal infection in children
- formerly Zygomycosis
- **Host factor:**
  - Immunosuppression
  - Hematologic malignant neoplasm
  - Renal failure
  - Diabetes mellitus
  - Iron overload syndromes
- **Route of entry:** respiratory tract, skin
- **Inhalation of sporangiospores** → pulmonary infection



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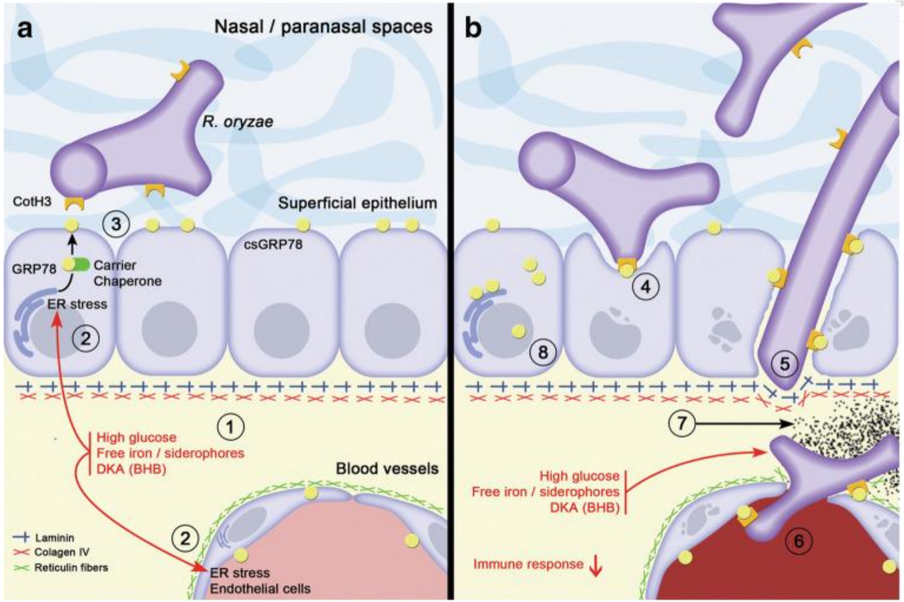
Pathogenesis of mucormycosis

- **Rhino-orbital/cerebral, sino-pulmonary mucormycosis**
  - Inhalation of spores from the environment

- **Cutaneous mucormycosis**
  - Inoculation of Mucorales spores to skin/subcutaneous tissues following severe trauma or abrasions on the skin



Clinical manifestations of mucormycosis

- **Most common:**
  - Rhinocerebral infections
  - Pulmonary infection
  - Other manifestations: GI tract, cutaneous, disseminated infection (CNS involvement)

- **Characteristic features:**
  - **Vascular invasion** → thrombosis, infarction, necrosis of surrounding tissue
- Rapid onset of necrotic lesion → need aggressive medical and surgical treatment

Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421.

Rhinocerebral mucormycosis

- Begins as necrotic lesion in paranasal sinuses → spread to orbit, face, palate and brain
- **Host factors:** neutropenia, organ transplantation, diabetic ketoacidosis, iron overload
- **Clinical presentations:**
  - Initial: nasal or sinus congestion, pain, epitaxis along with fever
  - Characteristic signs: black necrotic lesions on the nasal or palatine mucosa
  - Induration, necrosis and edema into periorbital or perinasal tissues
  - Ptosis, proptosis, loss of vision
  - Spread into cavernous sinus, brain → brain abscess

Figure 1: Cutaneous and rhino-orbito-cerebral mucormycosis

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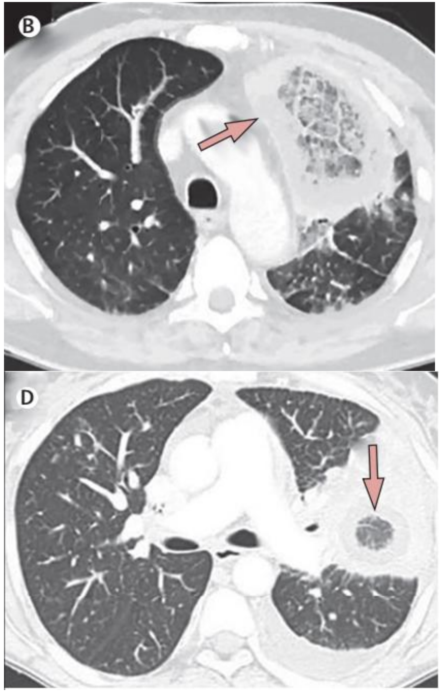
Pulmonary mucormycosis

- **Host factors:**
  - Severely neutropenic and severely immunocompromised children
  - HSCT → Graft-versus-host disease
- **Clinical presentations:**
  - Similar to other angioinvasive molds (*Aspergillus* spp.)
  - Pleuritic chest pain, cough, fever, hemoptysis
- **Radiologic signs:**
  - Nodular lesion
  - Cavitation
  - **Reversed halo sign** (**Inversed halo or atoll sign**):
    - Nodule with central ground-glass attenuation encircled by ring of denser consolidation



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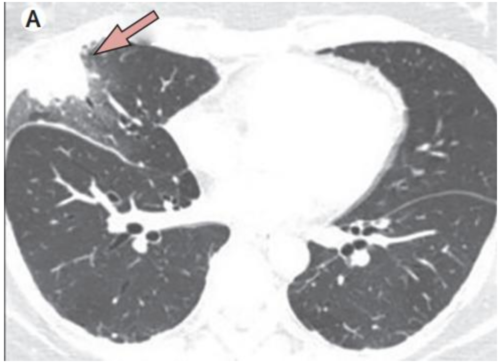
Imaging



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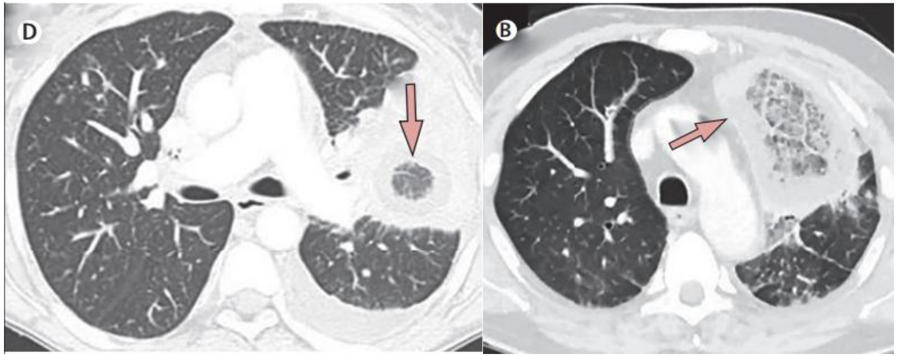

Halo sign on CT

- Ring of ground glass opacity surrounding a nodular infiltrate
- Typical of IPA

Reversed halo sign on CT

- Area of ground glass opacity surrounded by a ring of consolidation
- Found in pulmonary mucormycosis

Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421

Other manifestations of mucormycosis

	GI mucormycosis	Cutaneous mucormycosis	Disseminated infection
Clinical	- Not specific - Abdominal pain, hematemesis, GI bleeding - GI perforation, perirenal abscess, renal infarction	- History of trauma (minor) - Extensive necrotic cutaneous infections  - Site of infection: surgical site, burn wound, catheter site - Necrotizing cellulitis	- Follow primary site of mucormycosis; pulmonary, cutaneous - Focal cerebral infection - CNS mucormycosis → consider in severely immunocompromised with systemic involvement
Host factors	- Malnutrition - Severely immunocompromised	- Premature infants - Leukemia with contaminated medical devices, adhesive products	- Severely neutropenia

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Diagnosis of mucormycosis (1)

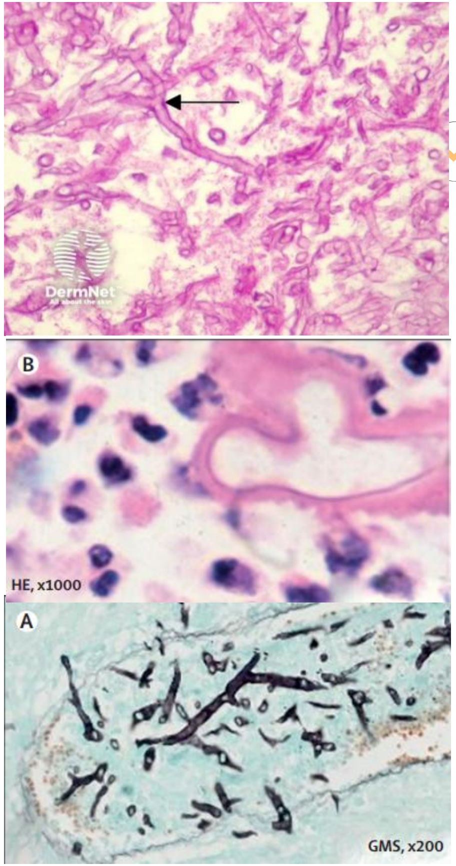
- Imaging
- Histopathology
- Culture and microscopy
- Susceptibility testing
- Molecular-based methods for direct detection
- Genus and species identification

Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421.

Diagnosis of mucormycosis (2)

Histopathology in mucormycosis

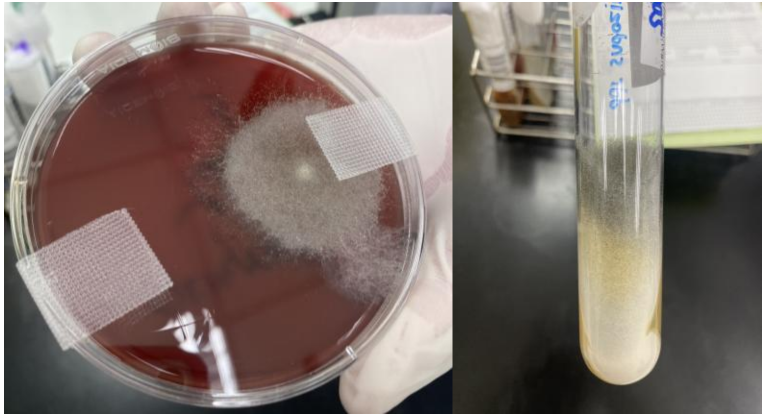
- Direct microscopy allows a rapid presumptive diagnosis of mucormycosis
- Hyphae of Mucorales
- Variable width (6-25 µm) vs 3-12 µm in Aspergillus
- Non-septate or pauci-septate
- Irregular, ribbon-like appearance
- Angle of branching is variable and includes wide angle (90 degrees) bifurcations
- Allow differentiation between hyphae of *Aspergillus* spp. vs Mucormycosis



Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421.

Diagnosis of mucormycosis (3)

- **Culture and microscopy** → low sensitivity
    - Recommended for genus and species identification, and for antifungal susceptibility testing



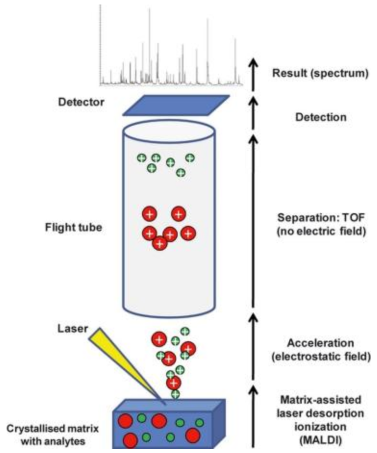
Cotton-candy like colony
from *Rhizopus* spp.

Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421.

Diagnosis of mucormycosis (4)

- **Molecular-based methods for direct detection**
  - Fresh material is preferred over paraffin-embedded tissue because formalin damages DNA
    - 18S rRNA, 28S rRNA sequencing
    - Internal transcribed spacer (ITS) sequencing

- **Genus and species identification**
  - To guide the choice of the antifungal treatment
  - To improve epidemiological knowledge of the disease
  - Matrix assisted laser desorption ionisation time of flight (MALDI-TOF)



Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421.

Treatment of mucormycosis (1)

Surgical treatment + **Immediate anti-fungal treatment**

- **Surgical debridement with clean margins**
- 3 purposes:

(1) disease control
(2) histopathology
(3) microbiological diagnostics

- Early complete surgical treatment for mucormycosis whenever possible
- Resection or debridement should be repeated as required

- **Anti-fungal agents:**
  - **Liposomal amphotericin B**
  - Isavuconazole (Children > 18 yrs)
  - Posaconazole
    - Delayed release tablet and oral suspension: Age ≥ 13 yrs
    - Powder: Age ≥ 2 yrs and BW ≤ 40 kg

*Avoid Amphotericin B deoxycholate due to toxicity

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Treatment of mucormycosis (2)

Conditions	Anti-fungal agents	Dosage and comments	Age approved
Primary therapy			
Empirical treatment	Liposomal amphotericin B	5–10 mg/kg per day from day 1	
If brain involvement	Liposomal amphotericin B	10 mg/kg per day from day 1	
If SOT	Liposomal amphotericin B or amphotericin B lipid complex	10 mg/kg per day from day 1	
Salvage therapy			
	Posaconazole	• Delayed-release tablets	≥ 2 year of age and BW > 40 kg
		• Delated-release oral suspension:	≥ 2 year of age and BW < 40 kg
	Isavuconazole	• USFDA approves expanded use - Need TDM monitoring	IV: ≥ 1 year of age Capsule: ≥ 6 years of age and BW ≥ 16 kg)

SOT; Solid Organ Transplant, TDM; Therapeutic drug monitoring

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Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421.
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Treatment of mucormycosis (3)

- **Adjunctive treatment in mucormycosis**
  - Discontinue and reduce immunosuppressive therapy or iron chelation treatment (if possible)
  - Hyperbaric oxygen therapy
  - Echinocandins → increase host response, in combination therapy

- **Duration of treatment**
  - Unknown, depends on clinical response, extent of surgical excision, persistence of underlying immunosuppressive pression
  - In general, **weeks to months of therapy**, until resolution of signs and symptom, immune recovery

Cornely OA, et al. Lancet Infect Dis. 2019 Dec;19(12):e405-e421.
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Prevention

- Reducing underlying risk factors that lead to the development of mucormycosis
- Control of diabetes mellitus (DM)
- Use of iron chelating agents other than deferoxamine
- Limit exposure of immunocompromised patients to soil and plants, laminar air filtration
- Limit exposure to hospital construction
  - Reduce the occurrence of aerosolization of molds, include Mucorales
- Caution the use of nonsterile items
  - often contaminated with environmental molds on skin and mucous membranes
    - low birth weight infants
    - immunocompromised patients

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Overview of Fungi

Yeast

- **Candida**
- **Cryptococcus**
- **Trichosporon**
- **Malassezia**

Mold

Aseptate

- **Mucorales**
- *Rhizopus spp.*
- *Mucor spp.*
- *Rhizomucor spp.*
  Lichtheimia spp
- **Entomophthorales**
- Basidiobolus
- Conidiobolus

Septate

Hyaline

- **Aspergillus**
- **Fusarium**
- **Scedosporium**

Dematiaceous

- **Phaeohyphomycosis**
  - *Alternaria* spp., *Bipolaris* spp., *Curvularia* spp., *Lomentospora prolificans*
- **Chromoblastomycosis**
  - *Fonsecaea spp.*

Dimorphic

- **Histoplasma**
- **Talaromyces**
- **Blastomyces**
- **Coccidioides**

Pediatric Invasive Fungal Infections: A Resident's Clinical Guide

FUNGAL CLASSIFICATION & THERAPY SPECTRUM

Yeast (e.g., Candida, Cryptococcus)

Molds (Septate e.g., Aspergillus, Aseptate e.g., Mucorales)

Dimorphic fungi (e.g., Histoplasma)

Escalation of Antifungal Therapy

Prophylaxis (unlikely disease/high risk) → Preemptive (possible) → Empirical (probable/septic) → Targeted therapy (proven disease)

INVASIVE CANDIDIASIS (YEAST) - The Leading Cause of IFI Mortality

High-Risk Pediatric Populations
- ICU stays, abdeminal surgery (especially with anastomotic leaks), broad-spectrum antibiotics, TPN, low-birth-weight neonates.

Emerging Resistance
- *C. auris* is multi-resistant; *C. krusei* is intrinsically resistant to Fiuconazole; 20% of *C. lusitaniae* inactive against Amphotericin B.

Candiduria in Preterm Infants
- Surrogate marker for candidemia; requires full evaluation (blood cultures, LP, renal/brain imaging).

Preferred Treatment | **Echinocandins**
First-line for moderately to severely ill children beyond neonatal period.

INVASIVE ASPERGILLOSIS (SEPTATE MOLD)

Most Common Form: Invasive Pulmonary Aspergillosis (IPA):
- Usually *A. fumigatus*; affects AML, aplastic anemia, HSCT/SOT patients.

Diagnostic "Gold Standard" and Assays
- Tissue biopsy is gold standard. Galactomannan (GM) assay is specific marker (Positive if serum ≥0.5 or BAL fiuid ≥1.0).

Neonatal Exception
- High-dose Amphotericin B deoxycholate preferred over Voriconazole to avoid potential visual adverse effects.

Drug of Choice | **Voriconazole**
Primary treatment for IPA in children ≥2 years old; TDM strongly recommended.

MUCORMYCOSIS (ASEPTATE MOLD)

Highly Aggressive and Angioinvasive
- Life-threatening infection often manifesting as Rhinoceretiral or Pulmonary disease.

Radiologic Sign: The Reversed Halo
- Characteristic CT finding: nodule with central ground-glass ottenuation encircled by a ring of denser consolidation.

Prevention through Environmental Control
- Limit exposure to hospital construction, soil, non-sterile items (e.g., adhesive tape).

Primary Therapy | **Liposomal Amphotericin B**
Standard dosing 5–10 mg/kg per day; escalate to 10 mg/kg if brain involvement or SOT.

CLINICAL REFERENCE TABLES

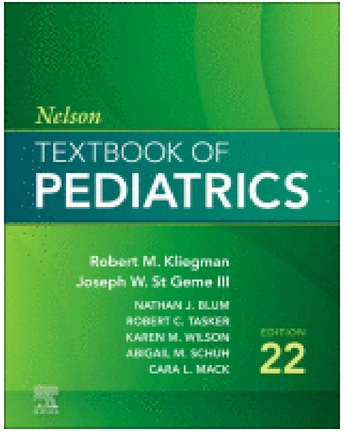
Pediatric Antifungal Dosing Reference

Drug	Infants Dosing	Older than 1 Year Dosing	Cautions
Amphotericin B deoxycholate	1 mg/kg/day	1 mg/kg/day	Preferred for neonates
Liposomal Amphotericin B	5 mg/kg/day	5 mg/kg/day	
Fluconazole	12 mg/kg/day	12 mg/kg load, then 6 mg/kg/day	Useful for UTIs (high urine conc.)
Voriconazole	Not studled/approved	8 mg/kg every 12 hr	For ≥2 yrs; TDM required
Micafungin	10 mg/kg/day	2-4 mg/kg/day	

Tissue Penetration (Tissue:Plasma Ratio)

Antifungal	CSF	Brain	Vitreous	Urine
Fluconazole	0.52–0.82	1.16–1.30	0.70	2.20–10.0
Voriconazole	0.38–0.68	1.20–1.90	0.40	0.01
Liposomal AmB	<0.01	0.03	0.03	Low
Echinocandins	0.00	0.10–0.20	<0.01	0.02–0.38

References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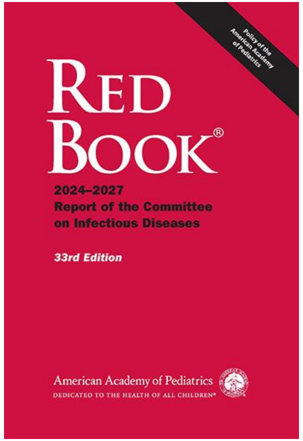
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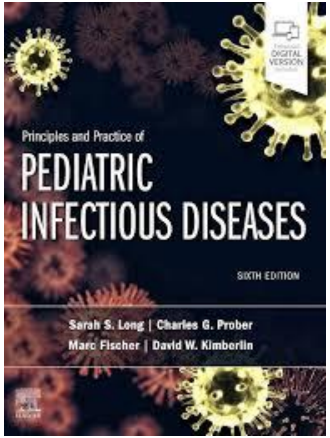
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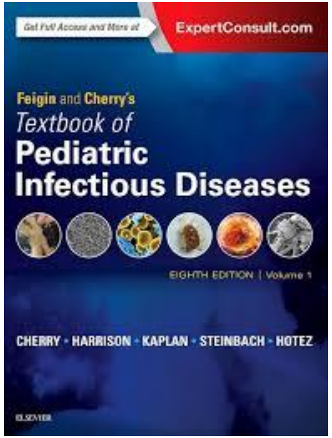
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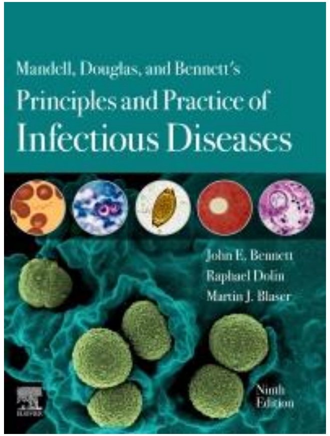
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Thank you

Q&A

Kahoot!